

Jaydipsinh Padhiyar

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Summary

Software Developer with a Master's in Applied Computer Science and hands-on experience across **full-stack** development, **cloud** platforms (**AWS**, **Azure**, **GCP**), and **DevOps** tooling. Skilled in building scalable **microservices**, **REST APIs**, and **CI/CD** pipelines with **Docker** and **Kubernetes**, delivering reliable, maintainable systems end to end.

Skills

Languages JavaScript, TypeScript, Python, C#, Go, Java
Front-End React, Redux, Next.js, HTML, CSS
Back-End Node.js, Express.js, Spring Boot, Django
APIs REST, OpenAPI, Swagger, GraphQL
Databases PostgreSQL, MongoDB, DynamoDB, MSSQL
Cloud AWS, Azure, GCP

DevOps Docker, Kubernetes, CI/CD, Jenkins, Terraform
Testing JUnit, Mockito, Jest, Cypress
Tools Git, GitHub, GitLab, Linux, Bash, Jira
Concepts Microservices, Cloud-Native, Agile, Performance Optimization, Security

Experience

Intern Support Consultant

Feb 2026 – Present

Nova Scotia Health Authority

Halifax, Canada

- Built **Excel & PowerBI dashboards** to track **deployment**, downtime planning, and metrics for the Health CIS rollout.
- Analyzed operational datasets and generated **data-driven reports** supporting **Deployment & Operations** planning.
- Coordinated **deployment schedules and readiness tracking** using structured planning sheets.
- Built a **Power Platform workflow** (Microsoft Forms, Power Automate, SharePoint, Lists) to automate document submission, **duplicate validation**, and centralized tracking for **operational forms**.
- Contributed to development of **Slow Down Survivor**, a **Godot-based** strategy simulation used to model clinic readiness planning and operational decision-making.
- Implemented **game logic and UI components using GDScript**, representing clinic metrics such as Slow Down, Staff Training, Waitlist Capacity, and Morale.

Software Developer

Jan 2025 – May 2025

Persuasive Computing Lab

- Developed full-stack web application using **Django REST APIs** and **React/Redux**, implementing requirements analysis, design documentation, and comprehensive testing strategies.
- Built the frontend with **Next.js 15 (App Router)**, **React 19**, and **TypeScript**, using nested layouts and dynamic routes (article, activation, password-reset flows) for a type-safe, modular architecture.
- Participated in daily **Agile stand-ups**, sprint planning, and peer **code reviews**, collaborating with cross-functional team members to ensure code quality and project alignment.
- Designed and implemented **RESTful APIs** for real-time data processing, quiz scoring, and content delivery with proper error handling and validation.
- Applied **object-oriented design principles** and created UML diagrams for system architecture.
- Integrated third-party SDKs (Tobii Pro SDK, Hugging Face) and handled complex inter-system communication.

Software Developer Intern

Jan 2023 – Apr 2023

Bhaskaracharya National Institute for Space Applications and Geo-informatics

Gandhinagar, India

- Engineered a road surface quality investigation system using **full-stack** technologies (**MERN**, **Django**, **OpenCV**) following structured SDLC methodology.
- Trained and integrated **deep learning models** (FasterRCNN, SSD, RetinaNet) via **Esri ArcGIS arcgis.learn**, serving **RetinaNet** through a **Django API** and cutting false positives around 15% via **threshold tuning** and image preprocessing.
- Developed and optimized **Python** data processing pipelines for large spatial datasets, reducing image retrieval time by 60% through multithreading and efficient algorithms.
- Participated in technical design reviews, presented progress to stakeholders, and documented software documents.
- Utilized **Jira** for task tracking, reported blockers in daily stand-ups, and aligned deliverables with project milestones.
- Performed comprehensive software **testing** including unit tests, integration tests, and validation of fixes.

Software Developer Trainee

Jul 2021 – Nov 2021

CHARUSAT University

Anand, India

- Developed and deployed a **cloud-native** Institute Management Portal on **Azure App Services** for scalable operations.
- Implemented **RESTful APIs** with best practices for scalability, security, and maintainability in cloud environments.
- Integrated **Azure CosmosDB** for scalable data management, optimizing schema and query performance.
- Applied **CI/CD** and cloud **deployment** strategies to streamline development and **deployment** workflows.
- Participated in code reviews, technical documentation, and knowledge-sharing sessions with team members.

Education

Dalhousie University

Masters in Applied Computer Science

courses: Advanced Concepts of Software Development, Advanced Web Development, Data Management, Warehousing & Analytics

Jan 2024 - May 2025

3.94/4.3 GPA

CHARUSAT University

Bachelors of Technology in Computer Engineering

courses: Data Structures, Analysis of Algorithms, Computer Networking, Artificial Intelligence, Databases, Theory of Computation
Operating Systems, Software Engineering, Digital Image Processing, Design of Language Processor

Jun 2019 - Apr 2023

4.0/4.3 GPA

Projects

Bucktuality – Anonymous Video Chat Platform

DotNET 10, Go, Python, Kubernetes, Docker, Redis, SQL Server, Kafka, SignalR, WebRTC, YARP 

- Built distributed **microservices** in **Golang** (Gin), **.NET 10**, and **Python** for real-time in an event-driven architecture.
- Developed anonymous video and text chat using **SignalR**, **WebRTC**, screen sharing, and matchmaking.
- Designed an event-driven architecture with **Apache Kafka** for analytics, notifications, and moderation.
- Built backend services with **Redis**, **SQL Server**, and **Entity Framework Core** for users, sessions, and reporting.
- Hosted Self-managed **kubeadm Kubernetes** cluster using **StatefulSets**, **Deployments**, **PVs**, **Secrets**, and health probes.
- Implemented an API Gateway using **YARP** for centralized routing and inter-service communication.

K8s Microservice

(GCP, Terraform, NodeJS, ExpressJS, Microservice, Kubernetes) 

- Built **CI/CD** pipelines with Google Cloud Build to automate deployments to GKE, reducing manual intervention and improving delivery speed.
- Automated GKE cluster provisioning using **Terraform**, enabling scalable and consistent infrastructure **deployment**.
- Developed **microservices** using **Node.js** and **Express.js** to facilitate RESTful API communication between containers.
- Configured **Kubernetes** YAML manifests for container orchestration and service exposure for scalable deployments.
- Developed and deployed containerized **microservices** on **Kubernetes**, optimizing resource usage and high availability.
- Improved developer workflows by reducing **deployment** friction through automated **CI/CD** pipelines.

Les Chat Aws

(AWS, EC2, VPC, RDS, CloudWatch, SNS, Elastic Load Balancing, GraphQL) 

- Designed a scalable, multi-tier **AWS** infrastructure using **CloudFormation** to ensure high availability and resilience.
- Implemented auto-scaling and self-healing cloud infrastructure for optimized resource utilization.
- Developed a secure **RDS PostgreSQL** database with multi-AZ **deployment** for fault tolerance.
- Deployed containerized applications using **Docker** to enable efficient cloud deployments.
- Configured comprehensive monitoring with **AWS CloudWatch** and set real-time alerts.
- Ensured high availability with a fault-tolerant network architecture using VPC, private subnets, and NAT Gateway.
- Automated deployments with scripts using infrastructure-as-code approach.

DalVacationHome

(ReactJS, NodeJS, Python, AWS, GCP, Docker, DynamoDB) 

- Architected and deployed a multi-cloud booking app on **AWS** and **Google Cloud**, ensuring scalability and uptime.
- Designed an Amazon Lex Chatbot with **AWS Lambda**-based booking logic for enhanced user interaction.
- Implemented an ETL process using **BigQuery** and **Google Natural Language API** for sentiment analysis.
- Automated deployments via **GitLab CI/CD**, implementing Infrastructure as Code with **CloudFormation**.
- Created booking and notifications using **AWS SNS**, **SQS**, and **GCP Pub/Sub** for real-time communication.

FurrEverHome

(SpringBoot, ReactJS, MySQL, Maven, Junit, Docker) 

- Built **CI/CD** pipeline with **Docker** and **GitLab**, reducing build and **deployment** times by 60% through automation.
- Used Test Driven Development (TDD) with JUnit to achieve 80% code coverage and enforce quality gates.
- Followed a structured **SDLC** approach from planning to **deployment** to meet all requirements.
- Deployed application on university VM with secure, scalable operations and 99.9% uptime using cloud best practices.

SOFABoot – Open Source Contribution (Merged PR)

(Java, JUnit 5, Mockito, Spring Boot/SOFABoot) 

- Authored unit tests for core components (startup/StartupReporter, compatibility/VerificationResultTests, util/SofaBootTestUtils) to harden startup metrics, compatibility checks, and environment utilities.
- Documented coverage in the PR (images), showing clear increases in line and branch coverage for the touched modules.
- Expanded edge-case validation for startup reporting, verification outcomes (success/failure states), and environment variable resolution to ensure deterministic behavior.
- Adopted **JUnit 5** + **Mockito** patterns for isolated, fast tests with concise assertions and clear failure messages.
- PR merged into the upstream **SOFABoot** codebase as an open-source contribution.